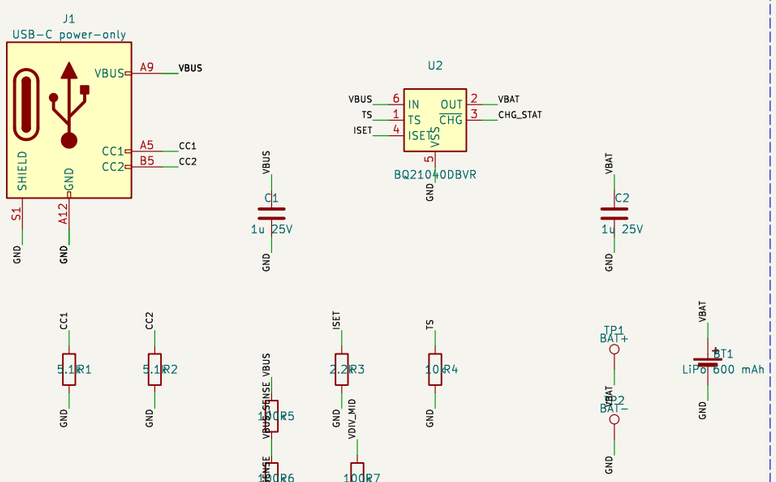
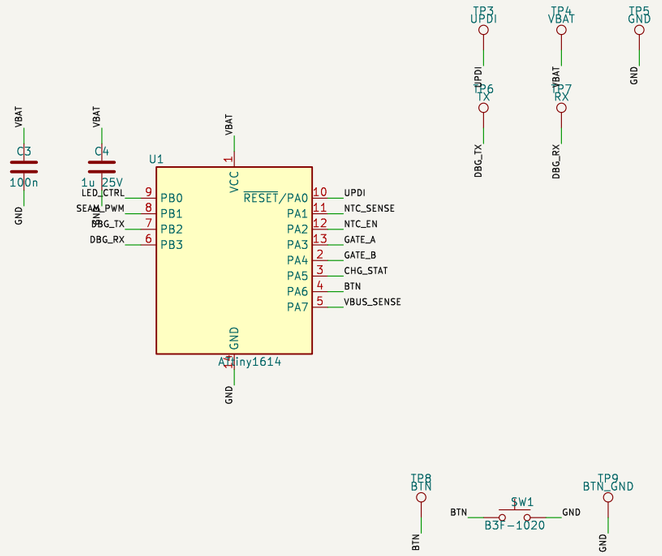
USB-C input, charger, cell

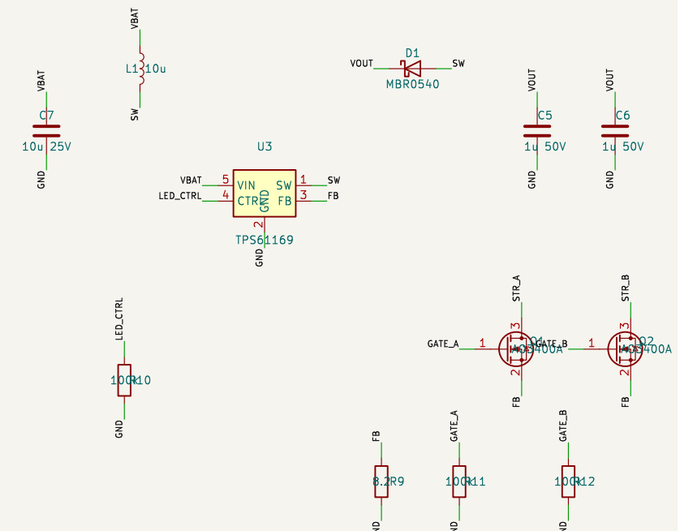


BQ21040: I_CHG = 540 / 2.2k = 245 mA (0.41C); 4.20 V; IN 30 V abs max, OVP 6.6 V

Microcontroller, button, programming

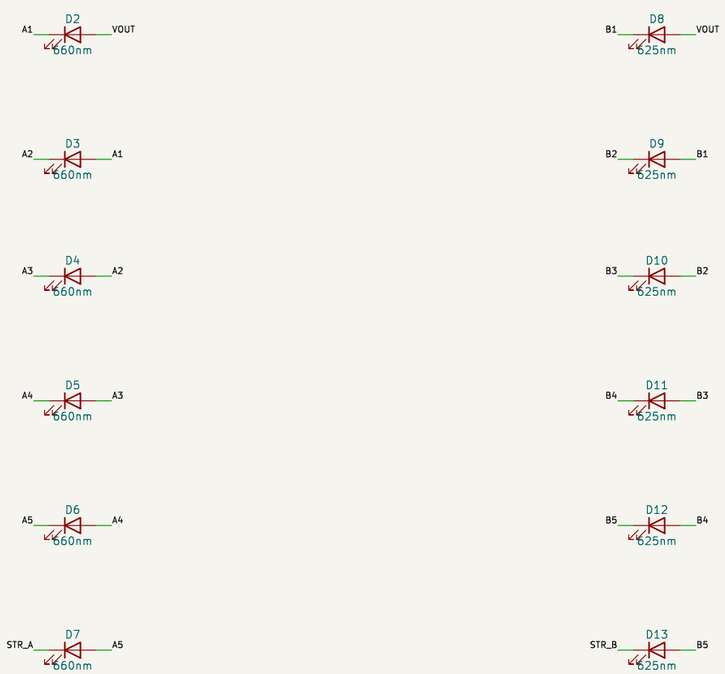


Boost LED driver and bank select

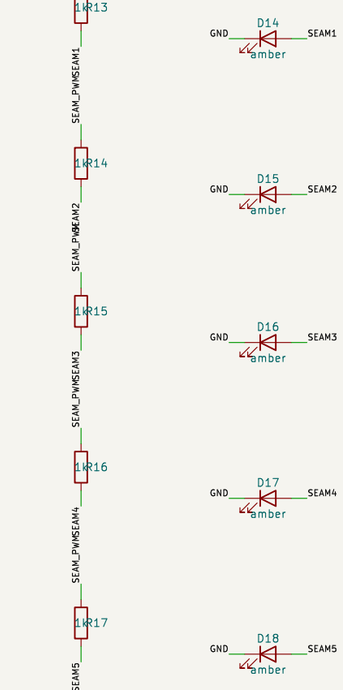


I_LED max = VREF / R9 = 204 mV / 8.2 = 24.9 mA (22.9–26.8); CTRL PWM 20 kHz

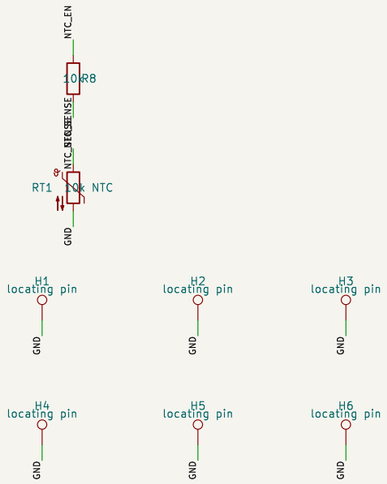
LED ring (PCB underside): string A 660 nm, string B 625 nm



Status seam



Temperature sense, locating pins



RT1 sits in the sole's 0.7 mm sensor window

LED ring and NTC on the PCB underside; ring positions from the CAD layout
Generated from ee/design.py (gate 3); nets by label
Personal prototype for bench evaluation — NOT a medical device
Personal prototype for bench evaluation — NOT a medical device

Sheet: /
File: vein_finder.kicad_sch

Title: Vein finder (organic) — main board

Size: A3	Date: 2026-10-12	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1